Discrete Structures Tutorial 3

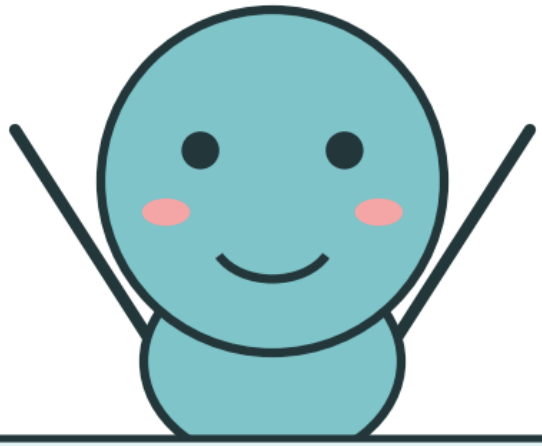
Ring Theory - Groups 11 & 12

TA: Meet Parekh

Before We Start...

Me, fresh from group theory

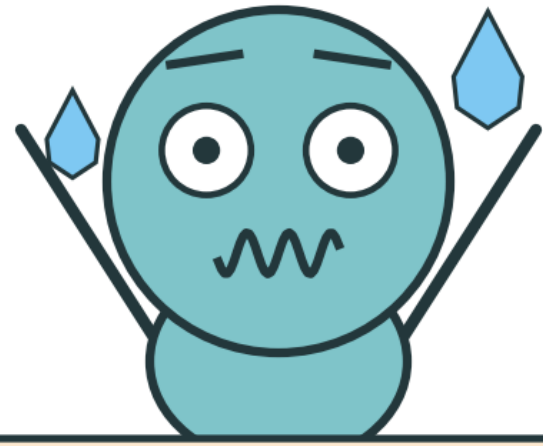
+



one set, one operation

Rings walk in

+ ×



a second operation!

...and they must get along: $a(b+c) = ab + ac$

From Groups to Rings

- **Group:** one set, **one** operation $*$. Closure, associativity, identity, inverses.
- **Ring:** one set R, **two** operations, $+$ and $\cdot$, that talk to each other.

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- **Group:** one set, **one** operation $*$. Closure, associativity, identity, inverses.
- **Ring:** one set R, **two** operations, $+$ and $\cdot$, that talk to each other.

A ring needs (the notes' R1–R8):

- $(R, +)$ is an **abelian group**: closed, commutative, associative, has 0, has $-a$.
- $\cdot$ is closed and **associative**.
- **Distributive laws** (the glue): $a(b+c) = ab + ac$ and $(b+c)a = ba + ca$.

In one line: a ring is a place where you can add, subtract and multiply. Division is not promised.

Ring Vocabulary

- **Commutative ring:** $ab = ba$ for all a, b.
- **Ring with unity:** there is an element, written 1, with $1 \cdot a = a \cdot 1 = a$ for all a (the ring's "multiplicative identity"). It is *called* 1 even when it does not look like the number 1.
- **Unit:** an element a that has an inverse, i.e. some b with $ab = ba = 1$.

Careful: unity vs unit

Unity = *the* element 1 of the ring. Exactly one per ring. (In $M_2(\mathbb{R})$ it is the identity matrix I.)

Unit = *any* element that has an inverse. A ring can have many.

1 is always a unit ($1 \cdot 1 = 1$). In $\mathbb{Z}$: one unity (1), two units (1 and -1).

Rings You Already Know

Ring	Commutative?	Unity?
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	yes	yes
$\mathbb{Z}_n$ (arithmetic mod n)	yes	yes (1)
$\mathbb{Z}[x]$ (integer polynomials)	yes	yes (1)
$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$	yes	yes (1)
$\mathbb{Z} \oplus \mathbb{Z}$ (componentwise)	yes	yes $((1, 1))$
$2\mathbb{Z}$ (even integers)	yes	**no**
$M_2(\mathbb{R})$ (2×2 real matrices)	**no**	yes (I)

Not a ring: $(\mathbb{N}, +, \cdot)$. There are no additive inverses, so $(R, +)$ is not a group.

Decoding Two Rows: $\mathbb{Z} \oplus \mathbb{Z}$ and $M_2(\mathbb{R})$

$\mathbb{Z} \oplus \mathbb{Z}$ = pairs (a, b) of integers, with everything done **one coordinate at a time**:

$$(a, b) + (c, d) = (a + c,\ b + d), \qquad (a, b) \cdot (c, d) = (ac,\ bd).$$

Zero $= (0, 0)$, unity $= (1, 1)$. Think: two copies of $\mathbb{Z}$ running side by side.

$M_2(\mathbb{R})$ = 2×2 real matrices. Addition is entrywise, multiplication is the usual matrix product (row times column). Zero = zero matrix, unity = identity matrix I.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{but} \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

So $AB \neq BA$ in general: $M_2(\mathbb{R})$ is **not commutative**.

Unity vs Unit: Side by Side

Ring	**Unity** (one element)	**Units** (all invertible elements)
$\mathbb{Z}$	1	$1,\ -1$
$\mathbb{Q},\ \mathbb{R},\ \mathbb{C}$	1	every nonzero number
$\mathbb{Z}_7$	1	$1, 2, 3, 4, 5, 6$ (all nonzero)
$\mathbb{Z}_{12}$	1	$1,\ 5,\ 7,\ 11$
$\mathbb{Z}[i]$	1	$1,\ -1,\ i,\ -i$
$\mathbb{Z} \oplus \mathbb{Z}$	$(1, 1)$	$(\pm 1, \pm 1)$: four of them
$M_2(\mathbb{R})$	I	matrices with det $\neq 0$
$2\mathbb{Z}$	**none**	(no 1, so nothing to invert to)

Rule of thumb: the unity is *one element* of the ring. The units are a *set* of elements. And 1 is always one of the units.

Does the Unity Have to Look Like 1?

No. "Unity" is defined by its *job*: an element e of the ring with $e \cdot a = a \cdot e = a$ for **every** a in that ring. We write it 1 out of habit, whatever it looks like.

Ring	**Its unity**	**It does nothing**
$\mathbb{Z} \oplus \mathbb{Z}$	the pair $(1, 1)$	$(1,1)(a,b) = (a,b)$
$M_2(\mathbb{R})$	the identity matrix I	$IA = AI = A$
$\mathcal{P}(\{a, b\})$	the whole set $\{a, b\}$	$\{a, b\} \cap X = X$

$\mathcal{P}(S)$: all subsets of S; "$X + Y$" = elements in exactly one of X, Y ($\triangle$); "$X \cdot Y$" = $X \cap Y$. Zero is $\varnothing$.

A Unity That Is Not 1 At All

Inside $\mathbb{Z}_6$ (clock with 6 hours), $E = \{0, 2, 4\}$ with the same $+$ and $\cdot$ is a ring on its own ($2 + 4 = 0$, $4 + 4 = 2$, $2 \cdot 4 \equiv 2$: it never leaves E). Multiply each element of E by 4 (mod 6):

x	0	2	4
$4 \cdot x$	0	$8 \equiv 2$	$16 \equiv 4$

4 changes nothing, so 4 **is the unity of** E**.**

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What to take from this

- $1 \notin E$. The unity of E is not the unity of $\mathbb{Z}_6$. "Unity" always means *of that ring*.
- Units are measured against E's own unity: $2 \cdot 2 = 4$, so 2 is a unit of E.
- A ring has only one unity, but a ring inside another ring can have a different one.
- Extreme case: in the one-element ring $\{0\}$ the unity is 0 (so $1 = 0$). That is why domains and fields demand $1 \neq 0$.

Quick Check: Is 2 a Unit?

A unit needs some b with $2 \cdot b = 1$.

Ring	Is 2 a unit?	Why
$\mathbb{Z}$	no	no integer b has $2b = 1$
$\mathbb{Q}$	yes	$b = \frac{1}{2}$
$\mathbb{Z}_7$	yes	$b = 4$, since $2 \cdot 4 = 8 \equiv 1$
$\mathbb{Z}_{12}$	no	$2b$ is always even, never $\equiv 1$

This is exactly the field test

A **field** is a ring where this test passes for *every* nonzero element, not just 2.

What a Ring Does *Not* Promise

- **Commutativity.** $M_2(\mathbb{R})$ is a ring, but $AB \neq BA$ in general.
- **A unity.** $2\mathbb{Z}$ (even integers) has no 1. A unity u would need $u \cdot 2 = 2$, so $u = 1$, which is odd.
- **Cancellation.** In $\mathbb{Z}_{12}$: $3 \cdot 4 = 0 = 3 \cdot 8$, yet $4 \neq 8$. So "$ab = ac \Rightarrow b = c$" can fail.

Two Rings That Behave Like Number Systems

- $\mathbb{Z}[i]$ (Gaussian integers): multiply like complex numbers, $i^2 = -1$.

  Example: $(1+i)(2-i) = 2 - i + 2i - i^2 = 3 + i$.
- $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$. Every nonzero element is invertible:

$$\frac{1}{a+b\sqrt{2}} = \frac{a-b\sqrt{2}}{a^2-2b^2}, \qquad a^2 - 2b^2 \neq 0 \text{ since } \sqrt{2} \text{ is irrational.}$$

So $\mathbb{Q}[\sqrt{2}]$ is even better than a ring. We'll name that in a few slides.

Handy Rules (Same in Every Ring)

For all a, b in any ring R:

- $a \cdot 0 = 0 \cdot a = 0$
- $a(-b) = (-a)b = -(ab)$
- $(-a)(-b) = ab$

Why $a \cdot 0 = 0$?

$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$. Cancel $a \cdot 0$ in the **additive** group. Done.

The proof only uses distributivity and the fact that $(R, +)$ is a group. (E5.1 in the notes asks you to prove all three.)

Zero Divisors

Definition (commutative ring R)

A nonzero $a \in R$ is a **zero divisor** if $ab = 0$ for some nonzero $b \in R$.

Quick test: if $a \neq 0$, $b \neq 0$ and $ab = 0$, then **both** a and b are zero divisors (each is the other's witness).

- In $\mathbb{Z}_{12}$: $3 \cdot 4 = 12 = 0$. So 3 and 4 are zero divisors.
- In $\mathbb{Z} \oplus \mathbb{Z}$: $(1, 0) \cdot (0, 1) = (0, 0)$. Zero divisors again.
- In $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$: none. $ab = 0$ forces $a = 0$ or $b = 0$.

Integral Domains

Definition

An **integral domain** is a commutative ring with unity $1 \neq 0$ and **no zero divisors**.

- **Yes:** $\mathbb{Z}$, $\mathbb{Z}[x]$, $\mathbb{Z}[i]$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ for p prime.
- **No:** $\mathbb{Z}_{12}$, $\mathbb{Z}_6$, $\mathbb{Z} \oplus \mathbb{Z}$ (zero divisors).

Why people love them: cancellation works. If $a \neq 0$ and $ab = ac$, then $a(b - c) = 0$, so $b - c = 0$, so $b = c$.

Fields

Definition

A **field** is a commutative ring with unity in which **every nonzero element is a unit**.

- **Yes:** $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ (p prime), $\mathbb{Q}[\sqrt{2}]$.
- **No:** $\mathbb{Z}$ (only ± 1 are units), $\mathbb{Z}[x]$, $\mathbb{Z}_{12}$.

Every field is an integral domain: if $xy = 0$ and $x \neq 0$, then $y = x^{-1}xy = x^{-1} \cdot 0 = 0$.

The converse is false: $\mathbb{Z}$ is a domain that is not a field.

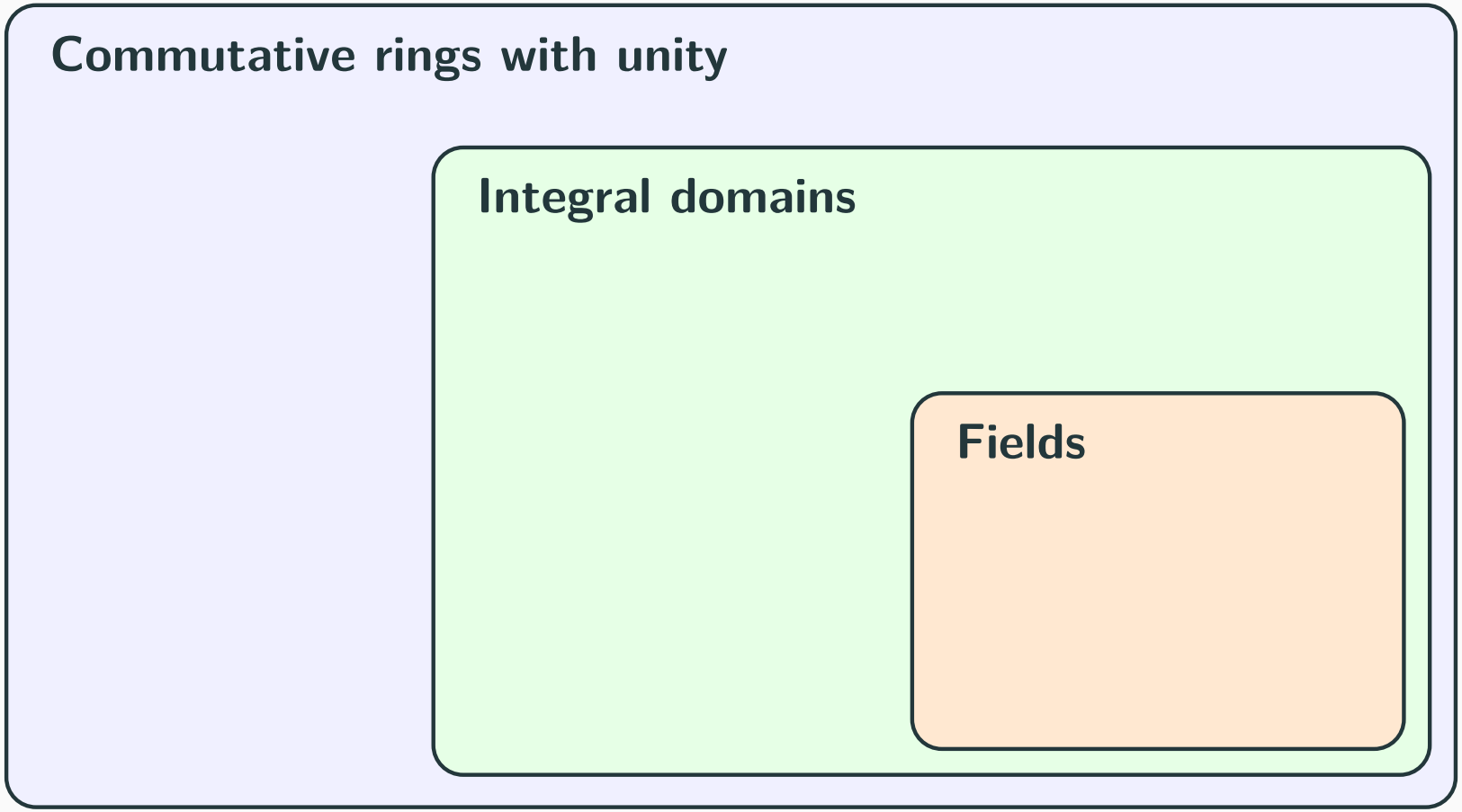
Why Does a Field Mention Unity and Units?

- **Unity = the target.** "b is the inverse of a" means $ab = 1$. No 1, nothing to aim at, so no inverses.
- **Units = the elements that can hit the target.** a is a unit if some b gives $ab = 1$.
- **Field = every nonzero element hits it.** That is exactly "you can divide by any nonzero a": $\dfrac{x}{a} = x \cdot a^{-1}$.

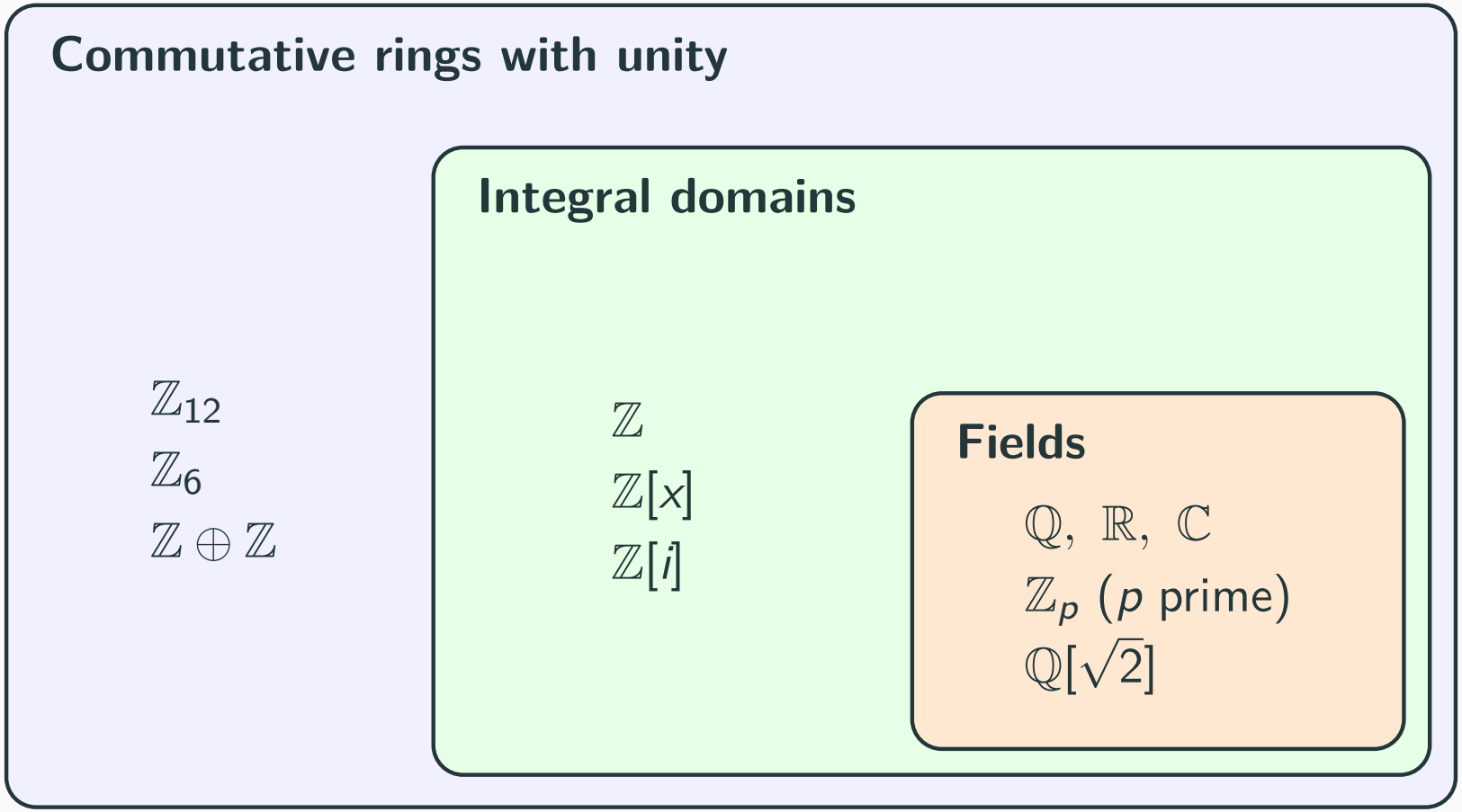
Why "nonzero"?

$0 \cdot b = 0 \neq 1$, so 0 is never a unit. (That uses $1 \neq 0$, which is why fields require it.)

The Big Picture: Who Lives Inside Whom



The Big Picture: Who Lives Inside Whom



Outside the whole picture: $2\mathbb{Z}$ (no unity) and $M_2(\mathbb{R})$ (not commutative).

Where Does a Ring Land? Three Questions

Ask these **in order**. The **first "no"** tells you which box of the nested picture the ring lands in.

Question	**If the answer is "no"**
Q1. Is it commutative, and does it have a 1?	just a ring (outside the nested boxes)
Q2. Does it have no zero divisors?	commutative ring with 1 only
Q3. Is every nonzero element a unit?	integral domain, but not a field
All three "yes"	**field**

Where Does a Ring Land? Examples

Same three questions, applied to rings we know.

Ring	Stops	Because	So it lands in
$M_2(\mathbb{R})$	Q1	$AB \neq BA$	just a ring (outside the nested boxes)
$2\mathbb{Z}$	Q1	it has no 1	just a ring (outside the nested boxes)
$\mathbb{Z}_{12}$	Q2	$3 \cdot 4 = 0$	commutative ring with 1 only
$\mathbb{Z} \oplus \mathbb{Z}$	Q2	$(1,0)(0,1) = (0,0)$	commutative ring with 1 only
$\mathbb{Z}$	Q3	2 has no inverse	integral domain, but not a field
$\mathbb{Q}$, $\mathbb{Z}_7$	none	every nonzero is a unit	**field**

Where This Is All Heading

Commutative ring with unity



Integral domain

no zero divisors



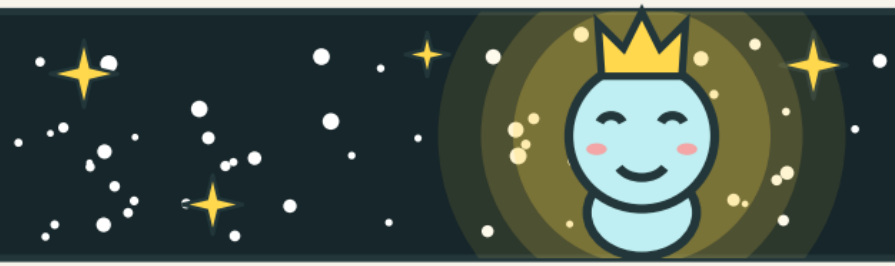
Field

every nonzero element is a unit



Finite integral domain

it was a field all along



Finite Integral Domain $\Rightarrow$ Field: Have vs Need

What we have (hypotheses)

D is a ring that is

- **finite**: only finitely many elements,
- **commutative** with unity $1 \neq 0$,
- with **no zero divisors**: $xy = 0 \Rightarrow x = 0$ or $y = 0$.

What we must prove (goal)

Every nonzero $a \in D$ has an inverse: some $b \in D$ with $ab = 1$.

Then D is a field.

We are not told how to find b. We only have to show that it exists.

The Idea: Multiply Everything by a

$\mathbb{Z}_5$ **is a field.** Multiply every element by $a = 2$:

x	0	1	2	3	4
$2x$	0	2	4	1	3

Same five numbers, **reshuffled** (no repeats). So 1 appears: $2 \cdot 3 = 1$, and 3 is the inverse of 2.

$\mathbb{Z}_6$ **is *not* a field.** Multiply every element by $a = 2$:

x	0	1	2	3	4	5
$2x$	0	2	4	0	2	4

Repeats (0 and 3 both go to 0), so 1 is missed: 2 has no inverse.

Rule broken by $\mathbb{Z}_6$: no zero divisors

$2 \cdot 3 = 0$ with $2, 3 \neq 0$, and that is why the products repeat. (Only $1, 5$ are units in $\mathbb{Z}_6$.)

Finite Integral Domain $\Rightarrow$ Field: The Proof

Fix a nonzero $a \in D$. Define $f : D \to D$ by $f(x) = xa$ ("multiply by a").

1. **No repeats.** If $xa = ya$, then $(x - y)a = 0$. Since $a \neq 0$ and there are no zero divisors, $x - y = 0$, i.e. $x = y$.
2. **Everything is hit.** D has n elements and f produces n *different* outputs, so the outputs are all of D.
3. **So 1 is hit.** Some b has $ba = 1$. D is commutative, so $ab = 1$. Thus a is a unit. ■

Where each hypothesis was used: no zero divisors $\to$ Step 1. Finite $\to$ Step 2. Commutative $\to$ Step 3.

Why Both Hypotheses Are Needed

- **Drop "finite":** in $\mathbb{Z}$ (domain, infinite), $x \mapsto 2x$ never repeats, but it never hits 1. $\mathbb{Z}$ is not a field.
- **Drop "no zero divisors":** in $\mathbb{Z}_6$ (finite), $x \mapsto 2x$ repeats, and 1 is missed.

Takeaway

Finiteness turns "no repeats" into "everything is hit". No zero divisors gives "no repeats". Together they give inverses.

$\mathbb{Z}_n$: Field Exactly When n Is Prime

- $n = p$ prime: if $ab \equiv 0 \pmod{p}$ then $p \mid ab$, so $p \mid a$ or $p \mid b$. No zero divisors. $\mathbb{Z}_p$ is a finite integral domain, hence a **field**.
- $n = st$ with $1 < s, t < n$: then $s \cdot t = 0$ in $\mathbb{Z}_n$ with $s, t \neq 0$. Zero divisors, so not even a domain.

Takeaway

In $\mathbb{Z}_n$ every nonzero element is either a unit or a zero divisor, nothing in between. (You'll see it live in Problem 1.)

Why Care? Solving by Factoring

To solve $x^2 - 4x + 3 = 0$ we factor: $(x-1)(x-3) = 0$. Then we say: *a product is* 0 *only if one of the factors is* 0, so $x - 1 = 0$ or $x - 3 = 0$, i.e. $x = 1$ or $x = 3$.

The hidden assumption

"product $= 0 \Rightarrow$ some factor $= 0$" is exactly **no zero divisors**. It is true in $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$.

In an integral domain, solving by factoring is legal, and a polynomial of degree n has **at most** n **roots**.

Why Care? When It Breaks: $\mathbb{Z}_{12}$

Same equation $(x-1)(x-3)=0$, but now in $\mathbb{Z}_{12}$ (arithmetic mod 12):

x	$x-1$	$x-3$	$(x-1)(x-3) \bmod 12$	
1	0	10	0	
3	2	0	0	
7	6	4	$24 \equiv 0$	(no factor is 0!)
9	8	6	$48 \equiv 0$	(no factor is 0!)

Four roots for a degree-2 equation. The extra roots $7, 9$ are fake: they come from the zero-divisor pairs $6 \cdot 4 = 0$ and $8 \cdot 6 = 0$. In $\mathbb{Z}_{11}$ or $\mathbb{Z}_{13}$ (fields) there are only $x = 1, 3$.

Characteristic

Definition (ring with unity)

$\operatorname{char} R =$ the least $n \geq 1$ with $\underbrace{1 + 1 + \cdots + 1}_{n} = 0$. If there is no such n, then $\operatorname{char} R = 0$.

Think: *how many times do I add* 1 *to itself before it wraps around to* 0*?*

- $\operatorname{char} \mathbb{Z} = \operatorname{char} \mathbb{Q} = \operatorname{char} \mathbb{R} = 0$.
- $\operatorname{char} \mathbb{Z}_n = n$.
- $\operatorname{char} \mathbb{Z}_3[i] = 3$. ($\mathbb{Z}_3[i] = \{a + bi : a, b \in \mathbb{Z}_3\}$, a field with 9 elements.)

Characteristic of a Domain Is 0 or Prime: The Plan

Theorem

If R is an integral domain, then $\operatorname{char} R$ is 0 or a prime.

The claim is an **or**. To prove "A or B" we split into cases:

- **Case 1:** $\operatorname{char} R = 0$**.** The claim says "0 or prime". It is already true. **Nothing to prove.**
- **Case 2:** $\operatorname{char} R = n > 0$**.** Now we must show n **is prime**. Here $n \geq 2$, since $1 \neq 0$.

So all the work is in Case 2. The next slide does it.

Characteristic of a Domain: The Argument

Case 2. $\operatorname{char} R = n \geq 2$. Suppose n is *not* prime: $n = st$ with $1 < s, t < n$. (Here $s \cdot 1$ means $1 + 1 + \cdots + 1$, s ones.)

1. $s \cdot 1$ and $t \cdot 1$ are **nonzero**: each is a sum of fewer than n ones, and n is the *first* time that sum is 0.
2. But $(s \cdot 1)(t \cdot 1) = (st) \cdot 1 = n \cdot 1 = 0$ (distributivity).
3. Two nonzero elements with product 0 are zero divisors. A domain has none. **Contradiction.**

So n is prime. ■

Seeing it with $n = 6 = 2 \cdot 3$ in $\mathbb{Z}_6$

$2 \cdot 1 = 1 + 1 \neq 0$ and $3 \cdot 1 = 1 + 1 + 1 \neq 0$ (fewer than 6 ones), yet $(2 \cdot 1)(3 \cdot 1) = 6 \cdot 1 = 0$.

The Contrapositive: A Quick "Not a Domain" Test

Contrapositive (the same theorem, flipped): if char R is **neither** 0 **nor prime** (so it is a composite number), then R is **not** an integral domain.

- $\mathbb{Z}_6$ has characteristic 6. So it is not a domain, without searching. (Indeed $2 \cdot 3 = 0$.)
- Likewise $\mathbb{Z}_4$ (char 4) and $\mathbb{Z}_{12}$ (char 12).

Careful: the converse is false

Characteristic 0 or prime does **not** make a ring a domain. $\mathbb{Z} \oplus \mathbb{Z}$ (pairs of integers) has characteristic 0; $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ (pairs of 0/1, mod 2) has characteristic 2. In *both*, $(1, 0)$ and $(0, 1)$ are nonzero but $(1, 0)(0, 1) = (0, 0)$.

Same-looking pairs, different coordinates: integers in the first ring, mod-2 numbers in the second.

Subrings

Definition

A subset $A \subseteq R$ is a **subring** if A is itself a ring under the same $+$ and $\cdot$.

Subring test: A nonempty, and $a - b \in A$ and $ab \in A$ for all $a, b \in A$.

- $2\mathbb{Z} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ are all subrings of the next.
- $\mathbb{Z}[i] \subseteq \mathbb{C}$. $\{0, 2, 4\} \subseteq \mathbb{Z}_6$.

What Does "$a - b \in A$" Mean?

- A is a subset of R. "$a \in A$" reads "a **belongs to** A".
- The test: **pick any two elements** a, b **of** A. Compute $a - b$ inside R (that is, $a + (-b)$). The answer must **still be in** A.
- In words: *subtract two members, and you are still a member.*

Why this single test is enough

$a - a = 0 \in A$. Then $0 - a = -a \in A$ (negatives). Then $a - (-b) = a + b \in A$ (sums). So A contains 0, negatives and sums.

$A = 2\mathbb{Z}$: $6 - 4 = 2 \in 2\mathbb{Z}$. ✓ $A =$ odd numbers: $3 - 1 = 2 \notin A$. $\times$ (so odd numbers are not a subring.)

Ideals: Subrings That Absorb

Definition

A subring A of R is an **ideal** if for every $r \in R$ and $a \in A$, both ra and ar lie in A.

Plain words: multiply anything in A by *anything in* R (not just by things in A) and you stay in A. A **absorbs** R.

- **Ideal:** $n\mathbb{Z} \subseteq \mathbb{Z}$. In $\mathbb{R}[x]$: polynomials with constant term 0.
- **Subring, *not* an ideal:** $\mathbb{Z} \subseteq \mathbb{Q}$. Reason: $\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}$.

Short version: ideal = subring + absorbs outsiders.

What Does "Absorb" Mean?

Two different closure rules:

	Rule	**Who may multiply**
Subring (closed)	$a, a' \in A \Rightarrow aa' \in A$	**insiders** only
Ideal (absorbing)	$a \in A,\ r \in R \Rightarrow ra,\ ar \in A$	insider $\times$ **anyone in** R

Picture a sponge: once something is inside A, multiplying it by *anything* from the whole ring leaves it inside. Nothing escapes.

- Absorbing is the stronger rule: $A \subseteq R$, so absorbing $\Rightarrow$ closed.
- $\mathbb{Z} \subseteq \mathbb{Q}$ is closed ($\mathbb{Z} \cdot \mathbb{Z} \subseteq \mathbb{Z}$) but does not absorb: $\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}$.

An Ideal in Full Detail: $3\mathbb{Z}$ Inside $\mathbb{Z}$

$3\mathbb{Z} = \{\ldots, -6, -3, 0, 3, 6, 9, 12, \ldots\}$, the multiples of 3.

- **Subring?** $9 - 6 = 3 \in 3\mathbb{Z}$. $9 \cdot 6 = 54 = 3 \cdot 18 \in 3\mathbb{Z}$. ✓
- **Absorbs?** Take the insider 9. Multiply by *outsiders* (numbers not in $3\mathbb{Z}$): $2 \cdot 9 = 18$, $5 \cdot 9 = 45$, $(-7) \cdot 9 = -63$. All multiples of 3. ✓
- **In general:** $r \cdot (3k) = 3(rk)$, a multiple of 3 for every integer r.

So $3\mathbb{Z}$ is an ideal of $\mathbb{Z}$

It is a subring, and it absorbs: an insider times *any* integer is still a multiple of 3.

A Subring That Is Not an Ideal

Recall $\mathbb{Z} \oplus \mathbb{Z}$: pairs (a, b) of integers, with $+$ and $\cdot$ done coordinate by coordinate.

- $D = \{(a, a)\}$, the **diagonal**, is a subring:
  $(a, a) - (b, b) = (a - b, a - b) \in D$ and $(a, a)(b, b) = (ab, ab) \in D$.
- But it does **not absorb**: insider $(1, 1)$, outsider $(0, 5)$.
  $(1, 1) \cdot (0, 5) = (0, 5) \notin D$. So D is not an ideal.
- **Compare** $\mathbb{Z} \oplus \{0\} = \{(a, 0)\}$: $(a, 0) \cdot (c, d) = (ac, 0)$ stays inside for every (c, d). This one *is* an ideal.

Homomorphism in Plain Words

Let R and S be **any two rings** (not necessarily related, e.g. $\mathbb{Z}$ and $\mathbb{Z}_5$). A **ring homomorphism** $\varphi : R \to S$ sends elements of R to elements of S and respects both operations:

$$\varphi(a+b) = \varphi(a) + \varphi(b), \qquad \varphi(ab) = \varphi(a)\varphi(b).$$

Left side: $+, \cdot$ are done in R (add, then translate). Right side: they are done in S (translate, then add).

Example: reduction mod 5, $\varphi : \mathbb{Z} \to \mathbb{Z}_5$, $\varphi(n) = n \bmod 5$

Take 7 and 9 ($\varphi(7) = 2$, $\varphi(9) = 4$).
$\varphi(7+9) = \varphi(16) = 1$ and $\varphi(7) + \varphi(9) = 2 + 4 = 6 = 1$. ✓
$\varphi(7 \cdot 9) = \varphi(63) = 3$ and $\varphi(7)\varphi(9) = 2 \cdot 4 = 8 = 3$. ✓

Not a homomorphism: $\psi(n) = n^2$ on $\mathbb{Z}$. $\psi(1+1) = 4$ but $\psi(1) + \psi(1) = 2$.

Kernel in Plain Words

$$\ker \varphi = \{a \in R : \varphi(a) = 0\}$$

The **kernel** is everything the map sends to 0: the stuff it "wipes out".

- $\mathbb{Z} \to \mathbb{Z}_5$ (mod 5): kernel $= 5\mathbb{Z}$ (multiples of 5 become 0).
- $\mathbb{R}[x] \to \mathbb{R}$, $f \mapsto f(2)$ (plug in $x = 2$): kernel = polynomials with root 2, e.g. $x - 2$, $x^2 - 4$.
- $\mathbb{Z} \oplus \mathbb{Z} \to \mathbb{Z}$, $(a, b) \mapsto a$: kernel $= \{(0, b)\}$.

Why we care

$\varphi(a) = \varphi(b) \iff a - b \in \ker \varphi$. The kernel measures what the map **cannot tell apart**. In particular, φ is injective iff $\ker \varphi = \{0\}$.

Where Ideals Come From: Kernels

A **ring homomorphism** $\varphi : R \to S$ satisfies $\varphi(a+b) = \varphi(a) + \varphi(b)$ and $\varphi(ab) = \varphi(a)\varphi(b)$.
Its **kernel** is $\ker \varphi = \{a \in R : \varphi(a) = 0\}$.

Kernels are ideals

If $\varphi(a) = 0$ and $r \in R$, then $\varphi(ra) = \varphi(r)\varphi(a) = \varphi(r) \cdot 0 = 0$. So $ra \in \ker \varphi$: it **absorbs**.

- $\mathbb{Z} \to \mathbb{Z}_n$, $k \mapsto k \bmod n$. Kernel: $n\mathbb{Z}$.
- $\mathbb{R}[x] \to \mathbb{R}$, $f \mapsto f(0)$. Kernel: $\langle x \rangle$ (constant term 0).

Ideal = "the stuff that gets sent to 0". Every ideal is a kernel, of $R \to R/A$ (coming up).

Ideals Generated by Elements

In a commutative ring R with unity:

$$\langle a \rangle = \{ra : r \in R\} \quad \text{(principal ideal)}$$

$$\langle a_1, \ldots, a_n \rangle = \{r_1 a_1 + \cdots + r_n a_n : r_i \in R\}$$

- $\langle 3 \rangle = 3\mathbb{Z}$ in $\mathbb{Z}$. $\langle x \rangle$ in $\mathbb{R}[x]$: multiples of x.
- $\langle 2, x \rangle$ in $\mathbb{Z}[x]$: polynomials with **even constant term**.

Containment reverses divisibility

$\langle a \rangle \subseteq \langle b \rangle$ iff $b \mid a$. So in $\mathbb{Z}$: $\langle 6 \rangle \subseteq \langle 3 \rangle \subseteq \langle 1 \rangle = \mathbb{Z}$. **Bigger number, smaller ideal.**

Factor Rings

For an ideal A of R, the **factor ring** is the set of cosets

$$R/A = \{\, r + A : r \in R \,\},$$

with

$$(s + A) + (t + A) = (s + t) + A, \qquad (s + A)(t + A) = st + A.$$

- Same idea as G/N for groups, now with multiplication too.
- Example: $\mathbb{Z}/4\mathbb{Z} = \mathbb{Z}_4$. E.g.
  $(2 + 4\mathbb{Z})(3 + 4\mathbb{Z}) = 6 + 4\mathbb{Z} = 2 + 4\mathbb{Z}$.
- The zero of R/A is $0 + A = A$. Everything in A becomes 0.

Why Must A Be an Ideal?

The product must not depend on which representative we pick. Suppose $s' = s + a$ and $t' = t + b$ with $a, b \in A$. Then

$$s't' = (s+a)(t+b) = st + \underbrace{sb + at + ab}_{\text{must lie in } A}.$$

- sb and at: an element of A times an **arbitrary** element of R.
- That is exactly the absorbing rule.

So

The absorbing condition is precisely what makes multiplication of cosets well defined. No absorbing, no factor ring.

Factor Rings: "Mod Out" = Impose a Relation

Taking $R/\langle g \rangle$ means: **declare** $g = 0$ and see what survives.

What "declare 0" means (clock picture)

On a 12-hour clock we agree $12 = 0$, so $24, 36, \ldots$ are 0 too. Declaring $g = 0$ forces every multiple rg to be 0 as well. So the declared-zero set must absorb: it must be an ideal.

- $\mathbb{Z}/\langle n \rangle$: declare $n = 0$. You get $\mathbb{Z}_n$.
- $\mathbb{R}[x]/\langle x^2 + 1 \rangle$: declare $x^2 = -1$. Then x behaves like i, and you get $\mathbb{C}$.
- $\mathbb{Z}[i]/\langle 2 - i \rangle$: declare $2 - i = 0$, i.e. $i = 2$. Also $5 = (2 - i)(2 + i) = 0$. You get $\mathbb{Z}_5$.

(Gallian Ch. 14 and the class notes work through all three.)

Prime Ideals

In this part, R is commutative with unity.

Definition

A **proper** ideal P of R is **prime** if $ab \in P \Rightarrow a \in P$ or $b \in P$.

- Mimics "$p \mid ab \Rightarrow p \mid a$ or $p \mid b$".
- $p\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$ when p is prime.
- **Proper** means $P \neq R$. (R itself is excluded.)

Maximal Ideals

In this part, R is commutative with unity.

Definition

A **proper** ideal M of R is **maximal** if whenever $M \subseteq B \subseteq R$ with B an ideal, then $B = M$ or $B = R$.

- In words: **nothing fits strictly between M and R.**
- Picture it as the top shelf just below R in the lattice of ideals.

The Payoff

Let R be commutative with unity and A an ideal of R.

Theorem

- R/A is an **integral domain** $\iff$ A is **prime**.
- R/A is a **field** $\iff$ A is **maximal**.

Prime gives a domain. Maximal gives a field.

- Field $\Rightarrow$ domain, so **maximal** $\Rightarrow$ **prime**.
- The converse fails (Problem 8 shows a prime ideal that is not maximal).

Sanity Check in $\mathbb{Z}$

- p prime: $\mathbb{Z}/p\mathbb{Z} = \mathbb{Z}_p$ is a field. So $p\mathbb{Z}$ is **maximal**, hence also prime.
- $4\mathbb{Z}$: $\mathbb{Z}/4\mathbb{Z} = \mathbb{Z}_4$ has $2 \cdot 2 = 0$, not a domain. So $4\mathbb{Z}$ is **not prime**.
  Directly: $2 \cdot 2 \in 4\mathbb{Z}$ but $2 \notin 4\mathbb{Z}$.
- $\{0\} = \langle 0 \rangle$: $\mathbb{Z}/\langle 0 \rangle \cong \mathbb{Z}$ is a domain but not a field. So $\langle 0 \rangle$ is **prime, not maximal**.

That's the whole theory. Now let's actually use it.

Problem 1 (Notes E5.2)

Find all units and all zero divisors of $\mathbb{Z}_{12}$.

Problem 1: Units

Claim: $a \in \mathbb{Z}_{12}$ is a unit $\iff \gcd(a, 12) = 1$.

- If $\gcd(a, 12) = 1$: then $ax + 12y = 1$ for some integers x, y, so $ax \equiv 1 \pmod{12}$.
- If $ab \equiv 1$: any common divisor of a and 12 divides $ab - 12k = 1$, so gcd $= 1$.

Numbers in $1, \ldots, 11$ coprime to $12 = 2^2 \cdot 3$: 1, 5, 7, 11.
Check: $5 \cdot 5 = 25 \equiv 1$, $7 \cdot 7 = 49 \equiv 1$, $11 \cdot 11 = 121 \equiv 1$.

Units of $\mathbb{Z}_{12}$

$\{1, 5, 7, 11\}$, and each one is its own inverse.

Problem 1: Zero Divisors

Claim: a nonzero a is a zero divisor $\iff \gcd(a, 12) > 1$.

- If $d = \gcd(a, 12) > 1$: then $a \cdot \frac{12}{d} = \frac{a}{d} \cdot 12 \equiv 0$, and $0 < \frac{12}{d} < 12$, so the partner is nonzero.
- A unit is never a zero divisor: $ab = 0 \Rightarrow b = a^{-1}ab = 0$.

a	2	3	4	6	8	9	10
partner b	6	4	3	2	3	4	6
ab	12	12	12	12	24	36	60

Zero divisors of $\mathbb{Z}_{12}$

$\{2, 3, 4, 6, 8, 9, 10\}$. 4 units + 7 zero divisors = 11 nonzero elements.

Quick Breather

Me, my whole life

$ab = 0 \Rightarrow a = 0$ **or** $b = 0$



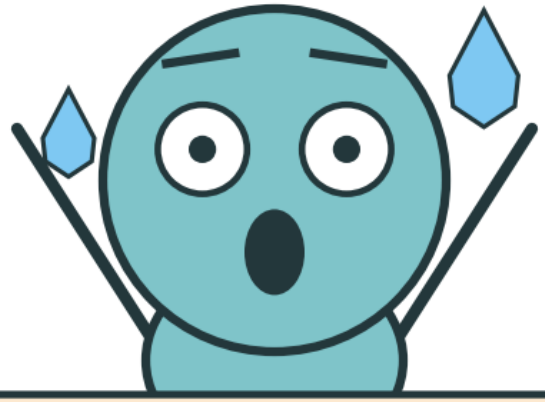
sounds totally safe

Then $\mathbb{Z}_{12}$ walks in

$3 \cdot 4 = 12$



$12 = 0$ **here**



$3 \cdot 4 = 0$**, both nonzero**

Two nonzero numbers, product zero: zero divisors exist

Problem 2 (Notes E5.3)

Show that $M_2(\mathbb{R})$ has zero divisors. Is it an integral domain?

Problem 2: Solution

(The notes define zero divisors for commutative rings. In $M_2(\mathbb{R})$ we mean nonzero A, B with $AB = 0$.)

Zero divisors. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \; B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} : \quad AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$A \neq 0$ and $B \neq 0$, so both are zero divisors.

Integral domain? No. Two independent reasons:

- It has zero divisors (above).
- It is not even commutative. With $C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ we get $AC = C$ but $CA = 0$.

Problem 3 (Notes E5.5)

A ring is Boolean if $a^2 = a$ for all a. Prove that a Boolean ring has characteristic 2 and is commutative.

Problem 3: Characteristic 2

Apply $x^2 = x$ to the element $a + a$:

$$a + a = (a + a)^2 = a^2 + a^2 + a^2 + a^2 = a + a + a + a.$$

Cancel $a + a$ in the additive group: $a + a = 0$ for every a.
So $2a = 0$ for all a. In particular $1 + 1 = 0$, i.e. $\operatorname{char} R = 2$. Also $-a = a$.

Example to keep in mind

$\mathcal{P}(\{1, 2, 3\})$ with $A + B = A \triangle B$ (symmetric difference) and $A \cdot B = A \cap B$.
$A \cap A = A$ (Boolean), and $A \triangle A = \varnothing$ (characteristic 2).

If R has no unity, "characteristic 2" just means $2a = 0$ for every a, which is what we proved.

Problem 3: Commutative

Apply $x^2 = x$ to $a + b$:

$$a + b = (a + b)^2 = a^2 + ab + ba + b^2 = a + ab + ba + b.$$

Cancel a and b (additive group): $ab + ba = 0$.

So $ab = -ba$. But $-x = x$ in a Boolean ring (previous slide), hence

$$ab = -ba = ba. \quad \blacksquare$$

Problem 4 (Notes E5.7)

Let R be a commutative ring of prime characteristic p. Prove that $(a+b)^p = a^p + b^p$ for all $a, b \in R$.

Problem 4: Solution (1/2)

R is commutative, so the binomial theorem holds:

$$(a+b)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^{p-k} b^k + b^p.$$

(Here $\binom{p}{k} x$ means x added to itself $\binom{p}{k}$ times.)

Claim: $p \mid \binom{p}{k}$ for $0 < k < p$.

- $k!\,(p-k)!\,\binom{p}{k} = p!$, and p divides $p!$.
- p is prime and every factor of $k!\,(p-k)!$ is $< p$, so $p \nmid k!\,(p-k)!$.
- Hence $p \mid \binom{p}{k}$.

Problem 4: Solution (2/2)

Write $\binom{p}{k} = p\,m$ with $m \in \mathbb{Z}$. For any $x \in R$:

$$\binom{p}{k} x = m\,(p \cdot x) = m\,\big((p \cdot 1)\,x\big) = m \cdot (0 \cdot x) = 0,$$

because $\operatorname{char} R = p$ means $p \cdot 1 = 0$.

So every middle term vanishes and $(a+b)^p = a^p + b^p$. ■

Sanity checks

$\mathbb{Z}_5$: $(1+1)^5 = 32 \equiv 2 = 1^5 + 1^5$. ✓

$\mathbb{Z}_4$ (characteristic 4, not prime): $(1+1)^2 = 0$ but $1^2 + 1^2 = 2$.

Fails, as it should.

(This is the "freshman's dream".)

Problem 5 (Notes E6.1)

Prove that every ideal of $\mathbb{Z}$ has the form $n\mathbb{Z}$ for some $n \geq 0$.

Problem 5: Solution (1/2)

Let I be an ideal of $\mathbb{Z}$.

- If $I = \{0\}$, then $I = 0\mathbb{Z}$. Done.
- Otherwise I has a nonzero element a, and $-a \in I$ too, so I has a **positive** element.

Let n be the **least positive element** of I. We show $I = n\mathbb{Z}$.

$n\mathbb{Z} \subseteq I$

$n \in I$ and I absorbs multiplication by integers, so $kn \in I$ for every $k \in \mathbb{Z}$.

Problem 5: Solution (2/2)

$I \subseteq n\mathbb{Z}$

Let $m \in I$. Divide: $m = qn + r$ with $0 \leq r < n$.

- $r = m - qn \in I$ (since $m \in I$ and $qn \in I$).
- $0 \leq r < n$ and n is the least positive element of I, so $r = 0$.

Hence $m = qn \in n\mathbb{Z}$.

So $I = n\mathbb{Z}$. ■

The engine was division with remainder. The same argument works in $F[x]$ for a field F, using degree instead of size.

Problem 6 (Notes E6.3)

Let R be commutative with unity. Prove that an ideal containing a unit is R. Prove that R is a field if and only if its only ideals are $\{0\}$ and R.

Problem 6: Part 1

Claim: if an ideal I contains a unit u, then $I = R$.

- $u^{-1} \in R$ and I absorbs, so $1 = u^{-1}u \in I$.
- For any $r \in R$: $r = r \cdot 1 \in I$.

So $I = R$. ■

Consequence

A **proper** ideal contains no units. (We'll use this again for prime and maximal ideals.)

Problem 6: Part 2

(We assume $R \neq \{0\}$, i.e. $1 \neq 0$.)

($\Rightarrow$) R a field, $I \neq \{0\}$ an ideal. Pick $a \neq 0$ in I. In a field, a is a unit. By Part 1, $I = R$. So the only ideals are $\{0\}$ and R.

($\Leftarrow$) Suppose the only ideals are $\{0\}$ and R. Let $a \neq 0$.

- $\langle a \rangle = \{ra : r \in R\}$ is an ideal containing $a \neq 0$, so $\langle a \rangle \neq \{0\}$.
- Therefore $\langle a \rangle = R$, so $1 \in \langle a \rangle$: $1 = ra$ for some r.
- So a is a unit.

Every nonzero element is a unit, so R is a field. ■

Problem 7 (Notes E6.6)

List all ideals of $\mathbb{Z}_{12}$. Which are prime? Which are maximal?

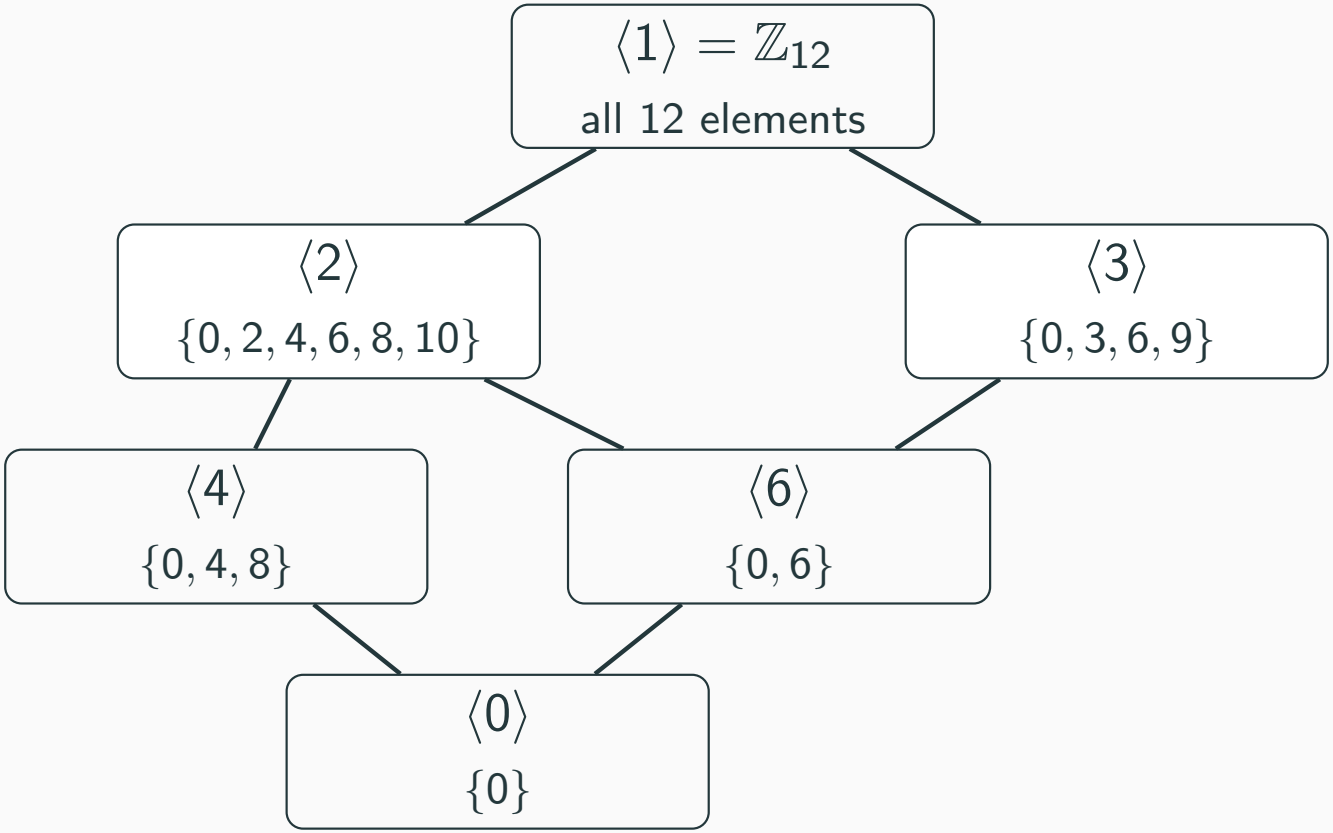
Problem 7: All the Ideals

An ideal of $\mathbb{Z}_{12}$ is, in particular, an additive subgroup of the cyclic group $\mathbb{Z}_{12}$. Its subgroups are generated by the divisors d of 12. And any $\langle d \rangle$ (a set of multiples of d) automatically absorbs.

$\langle 1 \rangle = \mathbb{Z}_{12}$	all 12 elements
$\langle 2 \rangle$	$\{0, 2, 4, 6, 8, 10\}$
$\langle 3 \rangle$	$\{0, 3, 6, 9\}$
$\langle 4 \rangle$	$\{0, 4, 8\}$
$\langle 6 \rangle$	$\{0, 6\}$
$\langle 12 \rangle = \langle 0 \rangle$	$\{0\}$

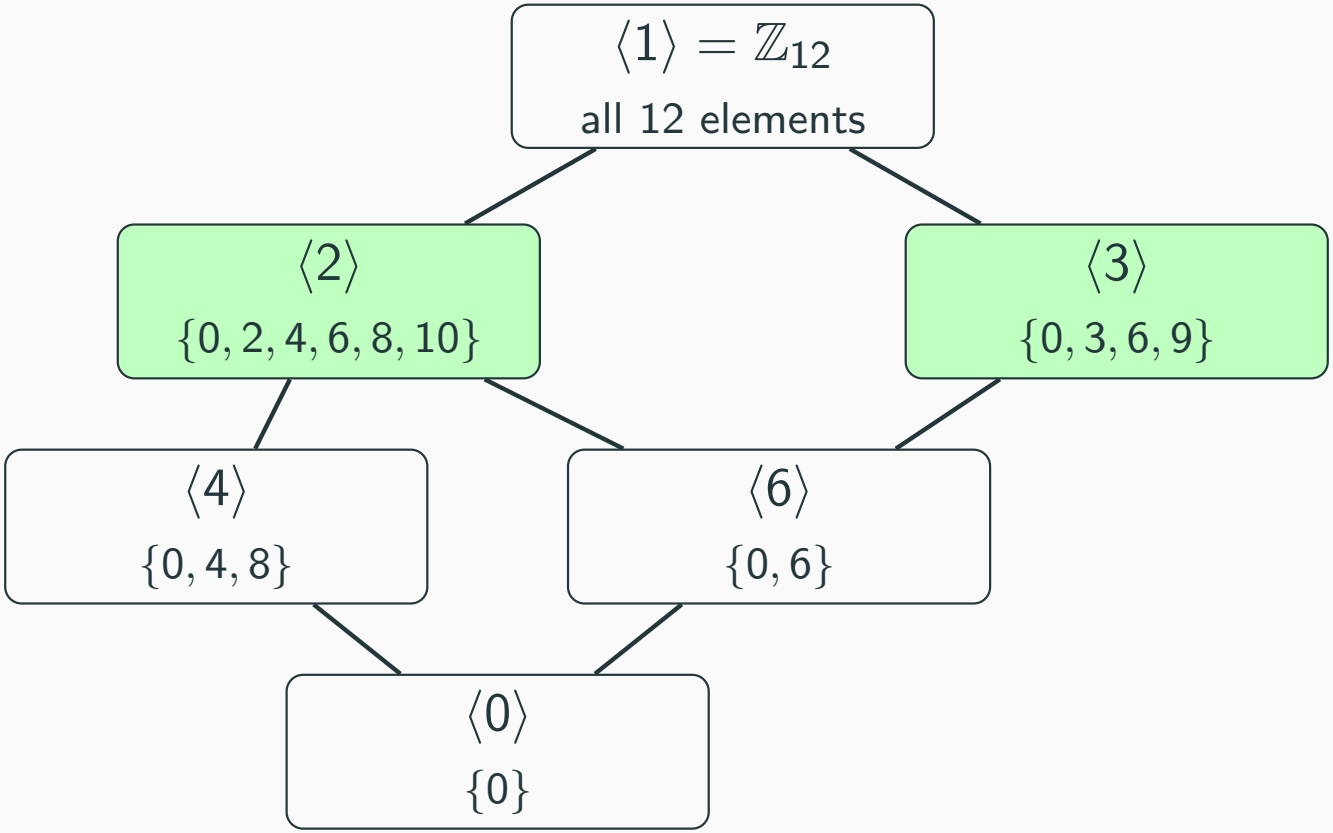
Exactly **six** ideals, one for each divisor of 12: $1, 2, 3, 4, 6, 12$.

Problem 7: Who Contains Whom



$\langle a \rangle \subseteq \langle b \rangle$ iff $b \mid a$. (Lines go downward = "is contained in".)

Problem 7: Maximal Ideals



Maximal = nothing strictly between it and $\mathbb{Z}_{12}$: exactly $\langle 2 \rangle$ and $\langle 3 \rangle$ (green).

Problem 7: Prime Ideals

Test each proper ideal A: is $\mathbb{Z}_{12}/A$ an integral domain?

Ideal	$\mathbb{Z}_{12}/A$	**Domain?**	**Witness**
$\langle 2\rangle$	$\mathbb{Z}_2$ (field)	yes	prime ✓
$\langle 3\rangle$	$\mathbb{Z}_3$ (field)	yes	prime ✓
$\langle 4\rangle$	$\mathbb{Z}_4$	no	$2\cdot 2 = 4 \in \langle 4\rangle$, but $2 \notin \langle 4\rangle$
$\langle 6\rangle$	$\mathbb{Z}_6$	no	$2\cdot 3 = 6 \in \langle 6\rangle$, but $2, 3 \notin \langle 6\rangle$
$\langle 0\rangle$	$\mathbb{Z}_{12}$	no	$3\cdot 4 = 0$, but $3, 4 \neq 0$

Answer

Prime = **maximal** $= \{\langle 2\rangle, \langle 3\rangle\}$. ($\langle 1\rangle = \mathbb{Z}_{12}$ is not proper.)

Problem 7: Why Prime = Maximal Here

- Maximal $\Rightarrow$ prime always (field $\Rightarrow$ domain).
- Prime $\Rightarrow$ maximal here because $\mathbb{Z}_{12}/A$ is **finite**. A finite integral domain is a field.

Pattern

In $\mathbb{Z}_n$, the prime ideals (= maximal ideals) are $\langle p \rangle$ for the **primes** p **dividing** n. For $n = 12$: $p = 2, 3$.

(Gallian Ch. 14, #6 asks for the maximal ideals of $\mathbb{Z}_8, \mathbb{Z}_{10}, \mathbb{Z}_{12}, \mathbb{Z}_n$. Try it.)

Hot Take Incoming

Every maximal ideal is prime.

Not every prime ideal is maximal.

Exhibit A: $\langle x \rangle$ in $\mathbb{Z}[x]$



Problem 8 (Notes E6.5)

In $\mathbb{Z}[x]$ prove that $\langle x \rangle$ is prime but not maximal.
(Hint: $\langle x \rangle \subsetneq \langle x, 2 \rangle \subsetneq \mathbb{Z}[x]$.)

Problem 8: $\langle x \rangle$ Is Prime

$\langle x \rangle = \{f \in \mathbb{Z}[x] : f(0) = 0\}$, the polynomials with constant term 0. It is **proper**: $1 \notin \langle x \rangle$.

Let $fg \in \langle x \rangle$. Then

$$f(0) \cdot g(0) = (fg)(0) = 0 \quad \text{in } \mathbb{Z}.$$

$\mathbb{Z}$ is an integral domain, so $f(0) = 0$ or $g(0) = 0$. That is, $f \in \langle x \rangle$ or $g \in \langle x \rangle$. ■

Slicker view

$f + \langle x \rangle$ and $f(0) + \langle x \rangle$ differ by a multiple of x, so they are the same coset. The cosets of $\langle x \rangle$ are therefore just the integers: $\mathbb{Z}[x]/\langle x \rangle$ looks like $\mathbb{Z}$ (we write $\cong \mathbb{Z}$), an integral domain. Prime $\iff$ quotient is a domain.

Problem 8: $\langle x \rangle$ Is Not Maximal

$\langle x, 2 \rangle = \{f : f(0) \text{ is even}\}$ (an ideal, see the notes).

- $2 \in \langle x, 2 \rangle$ but $2 \notin \langle x \rangle$ (constant term $2 \neq 0$). So $\langle x \rangle \subsetneq \langle x, 2 \rangle$.
- $1 \notin \langle x, 2 \rangle$ (constant term 1 is odd). So $\langle x, 2 \rangle \subsetneq \mathbb{Z}[x]$.

So $\langle x \rangle \subsetneq \langle x, 2 \rangle \subsetneq \mathbb{Z}[x]$: there is an ideal strictly between $\langle x \rangle$ and $\mathbb{Z}[x]$, so $\langle x \rangle$ is **not maximal**. ■

Consistent with the theorem

$\mathbb{Z}[x]/\langle x \rangle \cong \mathbb{Z}$ is a domain but not a field. And $\langle x, 2 \rangle$ *is* maximal: $\mathbb{Z}[x]/\langle x, 2 \rangle \cong \mathbb{Z}_2$.

Try These at Home (Build 1/2)

From the class notes:

- **E5.1** Prove $0a = a0 = 0$, $(-a)b = a(-b) = -(ab)$, $(-a)(-b) = ab$.
- **E5.4** The units of $\mathbb{Z}[i]$ are exactly $\pm 1, \pm i$.
- **E5.6** In a commutative ring with unity, the sum of two nilpotents is nilpotent.
- **E5.8** The characteristic of an integral domain is 0 or prime.
- **E6.2** If A, B are ideals, so are $A \cap B$ and $A + B$.
- **E6.4** A maximal ideal of a commutative ring with unity is prime.
- **E6.7** Kernels, and preimages of ideals, are ideals.
- **E6.8** $\langle 2, x \rangle$ is not a principal ideal of $\mathbb{Z}[x]$.

Try These at Home (Build 2/2)

From the class notes: E5.1, E5.4, E5.6, E5.8, E6.2, E6.4, E6.7, E6.8.

From Gallian (Contemporary Abstract Algebra):

- **Ch. 13, #4:** list the zero divisors of $\mathbb{Z}_{20}$. Notice the link with the units of $\mathbb{Z}_{20}$.
- **Ch. 13, #7:** a finite commutative ring with unity: every nonzero element is a zero divisor or a unit. What if we drop "finite"?
- **Ch. 13, #9:** $\mathbb{Z}[\sqrt{d}]$ is an integral domain.
- **Ch. 14, #5:** $\{a + bi : b \text{ even}\}$ is a subring of $\mathbb{Z}[i]$ but not an ideal.
- **Ch. 14, #6:** maximal ideals of $\mathbb{Z}_8, \mathbb{Z}_{10}, \mathbb{Z}_{12}, \mathbb{Z}_n$.
- **Ch. 14, #9:** $n\mathbb{Z}$ is a prime ideal iff n is prime.

Stuck? Bring it to TA hours.

Alright, We're Almost Done

Reject

ideals must absorb



Memorising "ideals must absorb" as a rule

Embrace



Asking once why *R/A* breaks without it

Rules you understand are rules you can rebuild

A Few Things Before You Go

- Read the actual class notes (Sections 5 and 6 are rings). Prof's proofs are highly testable.

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- Confirm the exact exam scope with the Professor, don't assume.
- Keep practicing: the "Try These at Home" list is a good start.

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- Keep practicing: the "Try These at Home" list is a good start.
- Use AI tools if you want, but verify the logic. No doubt is too small.

Let's Connect

TA: Meet Parekh

TA Hours:	Saturday, 3:00 PM – 4:00 PM
Location:	SERC, T-Hub 5th Floor
Email:	meet.parekh@students.iiit.ac.in
WhatsApp:	9327689404

TA: Meet Parekh

TA Hours:	Saturday, 3:00 PM – 4:00 PM
Location:	SERC, T-Hub 5th Floor
Email:	meet.parekh@students.iiit.ac.in
WhatsApp:	9327689404

- **Drop your doubt on WhatsApp before TA hours**: saves both our time, we go in ready.
- Not at my spot during TA hours? Message me, I'll come find you.

Thank You!

- Sorry if we rushed at the end!
- Drop me a DM with feedback (too fast? too slow? more memes?).



30-second anonymous feedback: Helpful? Not helpful? Anything else?

`https://forms.gle/Dhq9pSdo64vuGbye8`